configuration, let's just see it in action for now:

```
count=1
while [ $count -le 10 ]
do
  curl -o /dev/null -s -w "%{http_
code}\n" http://localhost
  count=$(( $count + 1 ))
done
```

This little *while* loop is making a call to the Apache Web server running locally, just for ten times, and displays the HTTP response code as the output. You will most likely see a few 'HTTP status code 200' which will be followed by 'HTTP status code 403'.

Configuring *mod_evasive*

Apache 2 provides several configuration scripts which are named *a2**, such as *a2query, a2enmod* and more. If you followed the steps above, you have already used them. These scripts primarily work by modifying *symlinks* inside the */etc/apache2* directory on the host.

When Apache starts up, it looks up the symlinks created in the */etc/apache2/mods-enabled/* directory and loads the module by reading the file names that match the **.load* patterns. For *mod_evasive,* the file will look like what follows:

```
LoadModule evasive20_module /usr/lib/
apache2/modules/mod_evasive20.so
```

The contents are self-explanatory. You will also see another *symlink* in the same directory that points to the *mod_evasive* configuration called *evasive.conf,* and the contents of this file will look like what's shown below:

```
<IfModule mod_evasive20.c>
    #DOSHashTableSize    3097
    #DOSPageCount        2
    #DOSSiteCount        50
    #DOSPageInterval     1
    #DOSSiteInterval     1
    #DOSBlockingPeriod   10

    #DOSEmailNotify      you@
yourdomain.com
    #DOSSystemCommand    "su - someuser
-c '/sbin/... %s ...'"
    #DOSLogDir           "/var/log/
mod_evasive"
</IfModule>
```

The official documentation (available at *https://github.com/jzdziarski/mod_evasive)* gives the following details about these parameters.

DOSHashTableSize: The hash table size defines the number of top-level nodes for each child's hash table. Increasing this number will provide faster performance by decreasing the number of iterations required to get to the record but consume more memory for table space. You should increase this if you have a busy Web server. The value you specify will automatically be tiered up to the next prime number in the primes list (see *mod_evasive.c* for a list of primes used).

DOSPageCount: This is the threshold for the number of requests for the same page (or URI), per page interval. Once the threshold for that interval has been exceeded, the IP address of the client will be added to the blocking list.

DOSSiteCount: This is the threshold for the total number of requests for any object by the same client on the same listener, per site interval. Once the threshold for that interval has been exceeded, the IP address of the client will be added to the blocking list.

DOSPageInterval: This is the interval for the page count threshold, which defaults to one-second intervals.

DOSSiteInterval: This is the interval for the site count threshold, which defaults to one-second intervals.

DOSBlockingPeriod: The blocking period is the amount of time (in seconds) that a client will be blocked for, if it has been added to the blocking list. During this time, all subsequent requests from the client will result in a 403 (Forbidden) error and the timer being reset (e.g., another 10 seconds). Since the timer is reset for every subsequent request, it is not necessary to have a long blocking period; in the event of a DoS attack, this timer will keep getting reset.

DOSEmailNotify: If this value is set, an email will be sent to the address specified whenever an IP address becomes blacklisted. A locking mechanism using */tmp* prevents continuous emails from being sent.

Note: Be sure MAILER is set correctly in *mod_evasive.c* (or *mod_evasive20.c*). The default is '/bin/mail -t %s' where *%s* is used to denote the destination email address set in the configuration. If you are running on Linux or some other operating system with a different type of mailer, you'll need to change this.

DOSSystemCommand: If this value is set, the system command specified will be executed whenever an IP address becomes blacklisted. This is designed to enable system calls to IP filters or other tools. A locking mechanism using */tmp* prevents continuous system calls. Use *%s* to denote the IP address of the blacklisted IP.

DOSLogDir: Choose an alternative temp directory. By default, */tmp* will be used for the locking mechanism, which opens up some security issues if your system is open to shell users *(http://security.lss.hr/index.php?page=details&ID=LSS-2005-01-01).*

In the event you have non-privileged shell users, you will need to create a directory writable only to the user Apache is running as (usually *root*), and then set this in your *httpd.conf.*

The parameters for your specific configuration may vary but the default settings are very strict. It blocks